Clinical SAS End to End PROJECT

Reviewer's Guide & Data Documentation

Phase II Oncology Study — Group B XP (Cisplatin & Capecitabine)

De-identified Dataset | Source: Project Data Sphere

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Programming	SAS studio
SDTM Standard	CDISC SDTMIG v3.4
ADaM Standard	CDISC ADaMIG v1.3
Validation Tool	Pinnacle 21 Community Edition
Data Source	Project Data Sphere — De-identified Oncology Dataset
Subjects (N)	436
Date	19-05-2026

This document is prepared as part of a personal learning and portfolio project.

1. About This Project

I built this SDTM and ADaM programming project independently as a fresher to develop hands-on experience with real clinical trial data. The dataset was sourced from the Project Data Sphere website, which provides de-identified oncology datasets for research and learning purposes.

Working with a real de-identified dataset — rather than synthetic data — gave me practical exposure to the kinds of data quality issues and inconsistencies that exist in actual clinical trial data. It also helped me understand how to handle and document these issues in a professional manner.

1.1 What I Built

#	Deliverable	Description
1	**SDTM Domains**	DM, AE, VS, DS, LB, EX — mapped from raw source to SDTM v3.4 standards
2	**ADaM Datasets**	ADSL,ADAE,ADCM and ADLB — built following ADaMIG v1.3
3	**Validation**	Ran Pinnacle 21 Community Edition on all domains; documented every error and warning
4	**TFL Templates**	

1.2 Why the Data Has Inconsistencies

Important: The source dataset was de-identified by Project Data Sphere before I downloaded it. The de-identification process involved date shifting, date masking, and removal of certain identifiers (site ID, country) and Treatment Group-A (Cetuximab + XP). This is the root cause of most date-related Pinnacle 21 errors seen in this project — it is not a programming error.

2. SDTM Domains and ADaM Datasets

2.1 SDTM Datasets

#	Domain	SDTM Class	Description	Records (N)	Dataset Used for ADaM
1	**DM**	Special Purpose	Demographics	436	ADSL
2	**AE**	Events	Adverse Events	7915	ADAE
3	**EX**	Interventions	Exposure	4515	ADSL / ADEX
4	**DS**	Events	Disposition	1303	ADSL
5	**LB**	Findings	Laboratory Results	91411	ADLB
6	**VS**	Findings	Vital Signs (Screening only)	3052	—
7	**CM**	Interventions	Concomitant medication	11944	ADCM

3.2 ADaM Datasets

S.no	Dataset	ADaM	Description	Source Domain(s)
1	**ADSL**	Subject Level	Subject-Level Analysis Dataset	DM, DS, EX
2	**ADLB**	BDS	Analysis Dataset — Laboratory	LB, ADSL
3	**ADAE**	OCCDS	Analysis Dataset — Adverse Events	AE, ADSL
4	**ADCM**	OCCDS	Analysis Dataset — Concomitant Medication	CM, ADSL

3. Pinnacle 21 Errors

Below is a complete record of every Pinnacle 21 error and warning generated when I validated my SDTM domains. For each error I have explained the root cause and what I did about it.

3.1 DM — Demographics

Errors kept as-is (source data limitation)

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD0002**	COUNTRY is blank for all records	**436**	Project Data Sphere removed country/region during de-identification. I cannot fill this in — kept blank.
2	**SD0002**	SITEID is blank for all records	**436**	Same reason as COUNTRY — removed during de-identification.
3	**SD1209**	RFXENDTC missing when RFXSTDTC is present	**4**	4 subjects have no exposure end date in the source — the data was removed during de-identification.
4	**SD1149**	ACTARMUD blank for all records	**436**	This is an expected (not required) variable. The source data does not contain actual arm description — kept blank.
5	**SD1149**	ARMNRS blank for all records	**436**	Same as above — expected variable, source data does not provide this. Kept blank.
6	—	RACE = 'OTHER' with no further detail	—	Some subjects have race recorded as 'OTHER' in the source with no further specification. Kept as-is.
7	**SD1478**	No LBOBXFL flag in LB for subject	**436**	LBOBXFL is permissible per SDTMIG v3.4 — not required. I did not derive it.
8	**SD1478**	No VSLOBXFL flag in VS for subject	**436**	Permissible per SDTMIG v3.5. Also, VS only has Screening visit data so last-observation flag cannot be derived.

Errors corrected

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD1334**	RFICDTC is after RFSTDTC	**217**	Date shifting in de-identified data caused this. fixed it by deriving RFSTDTC from RFXSTDTC (first exposure date) instead of using the informed consent date.
2	**SD1335**	RFICDTC is after RFXSTDTC	**217**	Same root cause. Fixed: RFXSTDTC re-derived from the earliest EXSTDTC value.
3	—	RFSTDTC is after RFENDTC	—	RFENDTC was derived from the discontinuation date but some values fell before RFSTDTC. Fixed:

				RFENDTC re-derived as the maximum of last discontinuation date, reference end date, and RFXSTDTC.
4	—	DTHFL has invalid value ('N')	—	originally coded DTHFL = 'N' for subjects not in the death dataset. Fixed: DTHFL = 'Y' only if died dataset shows death; otherwise kept blank (we cannot confirm alive status for missing subjects).

3.2 DS — Disposition

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**CT2005**	DSDECOD = 'OTHER' not in extensible codelist (DSCAT = DISPOSITION EVENT)	**35**	The source records the discontinuation reason as 'OTHER' with no further detail. Since this is an extensible codelist, I kept 'OTHER' in DSTERM and DSDECOD — no further information was available.
2	**SD1202**	DSSTDTC is after RFPENDTC	**173**	De-identified date inconsistency. Cannot correct without original data.
3	**SD1319**	DSSTDTC is before RFICDTC	**1**	Single record with de-identified date inconsistency. Cannot correct.

3.3 EX — Exposure

Error corrected

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD1352**	Duplicate records in EX domain	**78**	Found 78 duplicate records during programming. Fixed: sorted by USUBJID, EXTRT, EXCAT, EXROUTE, EXSTDTC, EXENDTC and removed exact duplicates using PROC SORT NODUPKEY.

Errors accepted (de-identification)

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD1202**	EXSTDTC is after RFPENDTC	**17**	De-identified date inconsistency.
2	**SD1206**	EXSTDTC is after RFXENDTC	**47**	De-identified date inconsistency.
3	**SD1446**	Exposure start date after latest Disposition date	**189**	De-identified date inconsistency.
4	**SD0082**	Exposure end date after latest Disposition date	**150**	De-identified date inconsistency.
5	**SD0021**	Missing End Time-Point value	**1,976**	Source contains missing end dates. Kept blank.
6	**SD0022**	Missing Start Time-Point value	**35**	Source contains missing start dates. Kept blank.

3.4 LB — Laboratory

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD1203**	LBDTC is after RFPENDTC	**5,006**	Large count because lab has ~25,000+ records and de-identified dates are inconsistent throughout. Cannot correct.

3.5 VS — Vital Signs

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD1083**	VSDY missing when VSDTC is present	**1**	Source does not provide vital sign collection dates. VSDTC and derived VSDY are both blank.
2	**SD1149**	VSDTC blank for all records	—	No collection date in source. Cannot derive.
3	**SD1149**	VSLOBXFL blank for all records	—	VS only contains Screening visit data — cannot derive a last-observation flag with only one timepoint.

3.6 CM — Concomitant Medications

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD0002**	CMTRT is blank (Required variable)	**4**	4 records have missing medication name in de-identified source. Kept blank.
2	**SD0013**	CMSTDTC is after CMENDTC	**3**	De-identified date inconsistency.
3	**SD0021**	Missing End Time-Point	**6**	Missing end dates in source.
4	**SD0022**	Missing Start Time-Point	**31**	Missing start dates in source.
5	**SD0031**	CMSTDTC/CMSTRF/CMSTRTPT missing when end info is present	**31**	De-identified data — start information not available.
6	**SD0035**	CMDOSU missing when CMDOSE is present	**249**	Dose unit not recorded in de-identified source.
7	**SD1202**	CMSTDTC after RFPENDTC	**980**	De-identified date inconsistency.
8	**SD1204**	CMENDTC after RFPENDTC	**1,032**	De-identified date inconsistency.

3.7 AE — Adverse Events

Two variables in AE needed special derivation logic because the source stored drug-specific information separately for Capecitabine and Cisplatin:

SDTM Variable	Source Variables	Derivation
AEREL	AECAPREL, AECISREL	If AECAPREL = 'YES' OR AECISREL = 'YES' → AEREL = 'RELATED'; if both = 'NO' → AEREL = 'NOT RELATED'
AEACN	CAPACT, CISACT	MEDICATION DISCONTINUED → DRUG WITHDRAWN | DOSE REDUCTION → DOSE REDUCED | MEDICATION DELAYED → DRUG INTERRUPTED | NONE → DOSE NOT CHANGED | NOT APPLICABLE → NOT APPLICABLE

#	Rule ID	Error Description	Count	Explanation / Action Taken
1	**SD0013**	AESTDTC is after AEENDTC	**2**	De-identified date inconsistency.
2	**SD0021**	Missing End Time-Point (AEENDTC)	**1,344**	Many AE end dates are missing in the de-identified source. Kept blank throughout.
3	**SD0022**	Missing Start Time-Point (AESTDTC)	**2**	2 records with missing AE start date.
4	**SD0080**	AE start date after latest Disposition date	**434**	De-identified date inconsistency.
5	**SD0090**	AESDTH not 'Y' when AEOUT = 'FATAL'	**3**	3 records where AE outcome is FATAL, but AESDTH is missing. This is a source data inconsistency could not correct.
6	**SD1202**	AESTDTC after RFPENDTC	**514**	De-identified date inconsistency.
7	**SD1204**	AEENDTC after RFPENDTC	**589**	De-identified date inconsistency.
8	**SD1332**	AEOUT = NOT RECOVERED but end date provided	**1,119**	Source data inconsistency due to de-identification.
9	**SD1449**	Missing MedDRA hierarchy codes (AEHLTCD, AEHLGTCD)	**7,915**	Source data does not contain higher-level MedDRA codes. All other MedDRA variables (AEDECOD, AEBODSYS etc.) are populated where available.
10	**SD0009**	No SAE qualifier = 'Y' when AE is Serious	**124**	Source has two datasets: ADE (7,948 records) and SAE (342 records). matched them using AEREFID = AESEQNO. Only matched records have SAE flags set to 'Y'; others left blank as SAE status unknown.

4. Error Summary

Domain	Main Error Type	Total Errors	Corrected	Kept (Justified)
DM	Missing vars, date issues	12	**4**	8 — source limitation
DS	Codelist, date issues	3	0	3 — de-identification
EX	Duplicates, date issues	7	**1**	6 — de-identification
LB	Date issues	1	0	1 — de-identification
VS	Missing dates and flags	3	0	3 — source limitation
CM	Missing values, date issues	8	0	8 — de-identification
AE	Dates, MedDRA, SAE mapping	10	**2 derivations**	8 — de-identification / structure
TOTAL		**44**	**5+**	**39 — All documented and justified**

5. What I Learned from This Project

This section summarises the key learnings I took away from building this project — things that I believe will directly help me in a clinical SAS programmer role.

5.1 SDTM Mapping Skills

1. How to map raw clinical data to SDTM using the correct controlled terminology, required variables, and SDTMIG naming conventions.
2. How to handle situations where source data does not contain all required or expected SDTM variables — and how to document those gaps.
1. The importance of SUPPQUAL datasets for variables that don't fit the standard domain structure.

5.2 Data Quality & Validation

2. How to use Pinnacle 21 Community Edition to validate SDTM domains and interpret the error/warning output.
3. The difference between errors that must be fixed vs. errors that are acceptable with a documented justification.
4. How de-identified datasets produce specific types of data inconsistencies (especially date-related) that are unavoidable.

5.3 ADaM Programming

5. How to build ADSL as the backbone dataset with key flags like SAFFL, DTHFL, and analysis population flags.
6. How to derive AVAL, BASE, CHG, PCHG, ANL01FL, and ABLFL in ADLB following BDS structure.

5.4 SAS Skills Developed

7. DATA step merge, PROC SQL joins, RETAIN statement, FIRST./LAST. logic, PROC TRANSPOSE, PROC SORT NODUPKEY.
8. Writing reusable macros and maintaining clean, commented SAS code.